

Wonjun Shin

其他

基本信息

姓名: Wonjun Shin

所属院校: 成均馆大学

地区: South Korea

学科领域

微电子科学与工程

材料物理

电子科学与技术

智能感知工程

材料科学与工程

集成电路设计与集成系统

电子信息材料

研究领域

薄膜材料与表面物理

微电子器件与制造

半导体与光电子材料

纳米电子与先进工艺

薄膜与表面技术

微纳传感器与MEMS

半导体器件物理与工艺

MEMS设计与制造

微纳电子与纳米技术

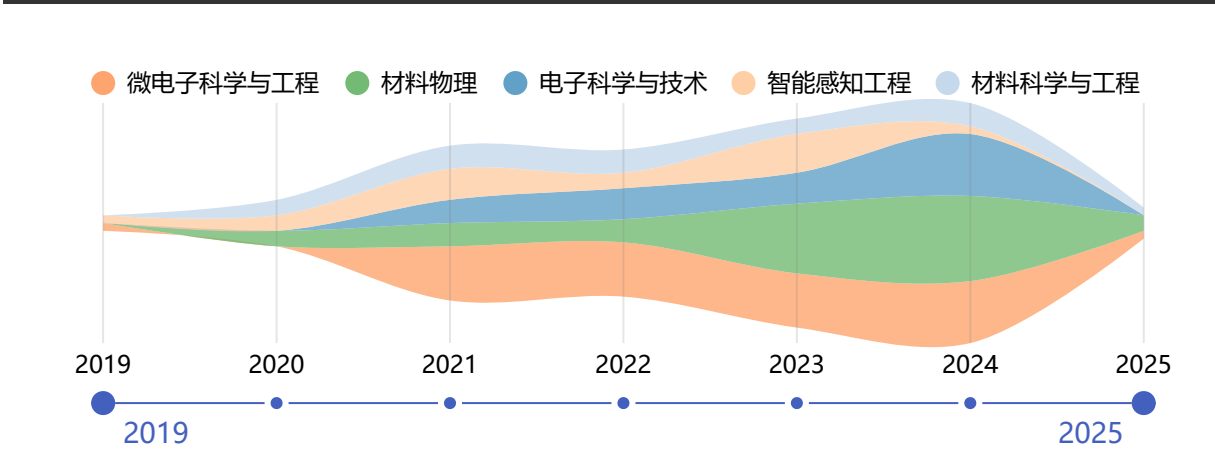
功能材料与智能材料

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个人简介

Wonjun Shin is a distinguished faculty member at Sungkyunkwan University, specializing in Semiconductor Devices and Low-Frequency Noise. With a robust academic profile, his research is highly cited, featuring an h-index of 21 and an i10-index of 41. His work primarily focuses on gas sensors, including FET-type and resistor-type sensors, with significant contributions to the understanding of low-frequency noise characteristics and the optimization of signal-to-noise ratios in semiconductor gas sensors. His collaborative efforts with leading institutions like Seoul National University and MIT further underscore his expertise in the field.

近年发表的论文



Does a large response suffice?: Thermally stable and low noise Si-doped IZO thin-film transistor-type gas sensors
SENSORS AND ACTUATORS B-CHEMICAL
Shin, Wonjun; Lee, Sunjin; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Sang Yeol; Lee, Sung-Tae
被引用次数: 0 2025

Stochastic behavior of random telegraph noise in ferroelectric devices: Impact of downscaling and mitigation strategies for neuromorphic applications
CHAOS SOLITONS & FRACTALS
Koo, Ryun-Han; Shin, Wonjun; Lee, Sung-Tae; Kwon, Daewoong; Lee, Jong-Ho
被引用次数: 0 2025

Does a large response suffice?: Thermally stable and low noise Si-doped IZO thin-film transistor-type gas sensors(vol 422 , 136498 , 2025)
SENSORS AND ACTUATORS B-CHEMICAL
Shin, Wonjun; Lee, Sunjin; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Sang Yeol; Lee, Sung-Tae
被引用次数: 0 2025

Examination of Ferroelectric Domain Dynamics in HZO Under Endurance Cycling Stress
IEEE ELECTRON DEVICE LETTERS
Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Kim, Seungwhan; Jung, Gyuweon; Lee, Sung-Tae; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho
被引用次数: 2 2024

Analog reservoir computing via ferroelectric mixed phase boundary transistors
NATURE COMMUNICATIONS
Kim, Jangsaeng; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Han, Chang-Hyeon; Kang, He Young; Yang, Tae Gyu; Goh, Youngin; Lee, Kilho; Ha, Daewon; Cheema, Suraj S.; Jeong, Jae Kyeong; Kwon, Daewoong
被引用次数: 1 2024

Charge-trap synaptic device with polycrystalline silicon channel for low power in-memory computing

SCIENTIFIC REPORTS

Park, Min-Kyu; Hwang, Joon; Kim, Soomin; Shin, Wonjun; Shim, Wonbo; Bae, Jong-Ho; Lee, Jong-Ho; Cho, Seongjae

被引用次数: 0

2024

Reliability of FET-type gas sensor with asymmetric air-gap micro-heater structure considering thermoelectric effect

SENSORS AND ACTUATORS B-CHEMICAL

Park, Jinwoo; Jung, Gyuweon; Shin, Wonjun; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Park, Min-Kyu; Kim, Jae-Joon; Lee, Jong -Ho

被引用次数: 1

2024

Proposition of optimal self-curing method in horizontal-floating gate FET-type gas sensors for reliability improvement

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Lee, Sung-Tae; Lee, Jong-Ho

被引用次数: 0

2024

Strain-Stress Impact on Ferroelectric Devices: A Multilayer Analysis and Optimization Strategy for Neural Networks

ACS MATERIALS LETTERS

Koo, Ryun-Han; Shin, Wonjun; Jung, Gyuweon; Kim, Jangsaeng; Lee, Sung-Tae; Im, Jiseong; Park, Sung-Ho; Ko, Jonghyun; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 0

2024

Photogating-based organic synapse electronics modulated by dielectric

ORGANIC ELECTRONICS

Kang, Seungme; Kim, Minseo; Yoo, Chansik; Lim, Byeong Min; Jang, Byung Chul; Shin, Wonjun; Lee, Hong-Sub; Yoo, Hocheon

被引用次数: 0

2024

Toward Optimized In-Memory Reinforcement Learning: Leveraging 1/f Noise of Synaptic Ferroelectric Field-Effect-Transistors for Efficient Exploration

ADVANCED INTELLIGENT SYSTEMS

Kim, Jangsaeng; Shin, Wonjun; Yim, Jiyong; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2024

Unveiled Influence of Sub-gap Density of States on Low-Frequency Noise in Si-Doped ZnSnO TFTs: Does Correlated Mobility Fluctuation Model Suffice?

ADVANCED ELECTRONIC MATERIALS

Shin, Wonjun; Lee, Ji Ye; Koo, Ryun-Han; Kim, Jangsaeng; Lee, Jong-Ho; Lee, Sang Yeol; Lee, Sung-Tae

被引用次数: 7

2024

Low-frequency noise analysis on asymmetric damage and self-recovery behaviors of ZnSnO thin-film transistors under hot carrier stress

DISCOVER NANO

Shin, Wonjun; Lee, Ji Ye; Kim, Jangsaeng; Lee, Sang Yeol; Lee, Sung-Tae

被引用次数: 0

2024

Robust 1/f Noise Unaffected by Program/Erase Cycling-Induced Damage in Ferroelectric Schottky Barrier FETs

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 1

2024

Vertical AND-Type Flash Synaptic Cell Stack for High-Density and Reliable Binary Neural Networks

IEEE ELECTRON DEVICE LETTERS

Kim, Jangsaeng; Im, Jiseong; Oh, Seongbin; Shin, Wonjun; Jung, Gyuweon; Lee, Sung-Tae; Lee, Jong-Ho

被引用次数: 2

2024

Proposition of Adaptive Read Bias: A Solution to Overcome Power and Scaling Limitations in Ferroelectric-Based Neuromorphic System

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Seungwhan; Im, Jiseong; Park, Sung-Ho; Ko, Jong Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2024

Toward Ideal Low-Frequency Noise in Monolayer CVD MoS₂ FETs: Influence of van der Waals Junctions and Sulfur Vacancy Management

ADVANCED SCIENCE

Shin, Wonjun; Byeon, Junsung; Koo, Ryun-Han; Lim, Jungmoon; Kang, Jung Hyeon; Jang, A-Rang; Lee, Jong-Ho; Kim, Jae-Joon; Cha, Seungnam; Pak, Sangyeon; Lee, Sung-Tae

被引用次数: 0

2024

Demonstration of In-Memory Biosignal Analysis: Novel High-Density and Low-Power 3D Flash Memory Array for Arrhythmia Detection

ADVANCED SCIENCE

Kim, Jangsaeng; Im, Jiseong; Shin, Wonjun; Lee, Soochang; Oh, Seongbin; Kwon, Dongseok; Jung, Gyuweon; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 1

2024

Polarization Pruning: Reliability Enhancement of Hafnia-Based Ferroelectric Devices for Memory and Neuromorphic Computing

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Jangsaeng; Yim, Jiyong; Ko, Jonghyun; Jung, Gyuweon; Im, Jiseong; Park, Sung-Ho; Kim, Jae-Joon; Cheema, Suraj S.; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 1

2024

Scaling Effects on Memory Characteristics of Ferroelectric Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Lee, Kitae; Yim, Jiyong; Shin, Wonjun; Kim, Sihyun; Kwon, Daewoong

被引用次数: 1

2024

Charge Transport Advancement in Anti-Ambipolar Transistors: Spatially Separating Layer Sandwiched between N-Type Metal Oxides and P-Type Small Molecules

ADVANCED FUNCTIONAL MATERIALS

Han, Youngmin; Lee, Subin; Kim, Minseo; Shin, Wonjun; Lee, Ho kyung; Koo, Ryun-Han; Lee, Sung-Tae; Kim, Chang-Hyun; Yoo, Hocheon

被引用次数: 4

2024

Random Number Generators and Spiking Neurons from Metal Oxide/Small Molecules Heterojunction N-Shape Switching Transistors

ADVANCED FUNCTIONAL MATERIALS

Seo, Juhung; Kang, Seungme; Kumar, Divake; Shin, Wonjun; Cho, Jinill; Kim, Taesung; Kim, Yeongkwon; Jang, Byung Chul; Trivedi, Amit R.; Yoo, Hocheon

被引用次数: 0

2024

Stochasticity in ferroelectric memory devices with different bottom electrode crystallinity

CHAOS SOLITONS & FRACTALS

Koo, Ryun-Han; Shin, Wonjun; Jung, Gyuweon; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 4

2024

Low-frequency noise of MoTe₂ transistor: effects on ambipolar carrier transport and CYTOP doping

DISCOVER NANO

Shin, Wonjun; Lee, Dong Hyun; Ko, Raksan; Koo, Ryun-Han; Yoo, Hocheon; Lee, Sung-Tae

被引用次数: 0

2024

NO₂ gas response improvement method by adopting oxygen vacancy controlled In₂O₃ double sensing layers

SOLID-STATE ELECTRONICS

Choi, Kangwook; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Lee, Chayoung; Kim, Donghee; Shin, Hunhee; Choi, Woo Young; Lee, Jong -Ho

被引用次数: 1

2024

Significant Reduction of 1/f Noise in Organic Thin-Film Transistors With Self-Assembled Monolayer: Considerations of Density-of-States

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jisuk; Park, Joon Hyung; Lee, Jong-Ho; Kim, Chang-Hyun; Lee, Sung-Tae

被引用次数: 1

2024

CMOS-Compatible Barometric Pressure Field-Effect Transistor-Type Sensor

IEEE ELECTRON DEVICE LETTERS

Lee, Chayoung; Shin, Wonjun; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

Optimizing Post-Metal Annealing Temperature Considering Different Resistive Switching Mechanisms in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 10

2023

Unraveling Threshold Voltage Instability in Ferroelectric Junctionless FETs Using Low-Frequency Noise Measurement With Base Bias

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2023

Comparative Analysis of n- and p-Type Ferroelectric Tunnel Junctions Through Understanding of Non-FE Resistance Switching

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Lee, Kyungmin; Park, Sung-Ho; Im, Jiseong; Ko, Jong-Hyun; Kim, Jeong-Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 6

2023

Ferroelectric Field-Effect Transistor Synaptic Device With Hafnium-Silicate Interlayer

IEEE ELECTRON DEVICE LETTERS

Kim, Sang Woo; Shin, Wonjun; Kim, Munhyeon; Kwon, Ki Ryun; Yim, Jiyong; Kim, Jeonghan; Han, Changhyeon; Jeong, Soi; Park, Eun Chan; You, Ji Won; Kim, Hyunwoo; Choi, Rino; Kwon, Daewoong

被引用次数: 3

2023

Effects of Sensor Platform Scaling on Signal-to-Noise Ratio in the Resistor-and Horizontal Floating-Gate FET-Type Gas Sensors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Jung, Gyuweon; Jeong, Yujeong; Lee, Sung-Tae; Lee, Jong-Ho

被引用次数: 1

2023

Annealing Ambient and Film Thickness Dependent NO₂ Response and 1/f Noise Characteristics of IGZO Resistor-Type Gas Sensors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Lee, Sung-Tae; Lee, Jong-Ho

被引用次数: 1

2023

Effect of Carrier Transport Process on Tunneling Electroresistance in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Dae Hwan; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

Unveiling Resistance Switching Mechanisms in Undoped HfO_x Ferroelectric Tunnel Junction Using Low-Frequency Noise Spectroscopy

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 9

2023

Channel-Length-Dependent Low-Frequency Noise Characteristics of Ferroelectric Junctionless Poly-Si Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kim, Sangwoo; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Deok-Hwang; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2023

Super-steep synapses based on positive feedback devices for reliable binary neural networks

APPLIED PHYSICS LETTERS

Kwon, Dongseok; Kim, Hyeongsu; Lee, Kyu-Ho; Hwang, Joon; Shin, Wonjun; Bae, Jong-Ho; Woo, Sung Yun; Lee, Jong-Ho

被引用次数: 2

2023

Recovery of off-state stress-induced damage in FET-type gas sensor using self-curing method

DISCOVER NANO

Shin, Wonjun; Jeong, Yujeong; Kim, Mingyu; Lee, Jungsoo; Koo, Ryun-Han; Hong, Seongbin; Jung, Gyuweon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 2

2023

Reconfigurable Manipulation of Oxygen Content on Metal Oxide Surfaces and Applications to Gas Sensing

ACS NANO

Jung, Gyuweon; Ju, Suyeon; Choi, Kangwook; Kim, Jaehyeon; Hong, Seongbin; Park, Jinwoo; Shin, Wonjun; Jeong, Yujeong; Han, Seungwu; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 20

2023

Energy Efficient Artificial Olfactory System with Integrated Sensing and Computing Capabilities for Food Spoilage Detection

ADVANCED SCIENCE

Jung, Gyuweon; Kim, Jaehyeon; Hong, Seongbin; Shin, Hunhee; Jeong, Yujeong; Shin, Wonjun; Kwon, Dongseok; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 8

2023

Self-Curable Synaptic Ferroelectric FET Arrays for Neuromorphic Convolutional Neural Network

ADVANCED SCIENCE

Shin, Wonjun; Im, Jiyong; Koo, Ryun-Han; Kim, Jaehyeon; Kwon, Ki-Ryun; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho; Kwon, Daewoong

被引用次数: 21

2023

In-Memory-Computed Low-Frequency Noise Spectroscopy for Selective Gas Detection Using a Reducible Metal Oxide

ADVANCED SCIENCE

Shin, Wonjun; Kim, Jaehyeon; Jung, Gyuweon; Ju, Suyeon; Park, Sung-Ho; Jeong, Yujeong; Hong, Seongbin; Koo, Ryun-Han; Yang, Yeongheon; Kim, Jae-Joon; Han, Seungwu; Lee, Jong-Ho

被引用次数: 6

2023

Macroscopic analysis and design of Si HFGFET gas sensor for sensitive gas detection

SOLID-STATE ELECTRONICS

Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Lee, Chayoung; Lee, Jong-Ho

被引用次数: 2

2023

Reconfigurable neuromorphic computing block through integration of flash synapse arrays and super-steep neurons

SCIENCE ADVANCES

Kwon, Dongseok; Woo, Sung Yun; Lee, Kyu-Ho; Hwang, Joon; Kim, Hyeongsu; Park, Sung-Ho; Shin, Wonjun; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 8

2023

Observation of interface trap reduction in fluoropolymer dielectric organic transistors by low-frequency noise spectroscopy

APPLIED PHYSICS LETTERS

Shin, Wonjun; Shin, Jihyun; Lee, Jong-Ho; Yoo, Hocheon; Lee, Sung-Tae

被引用次数: 2

2023

Effects of temperature and DC cycling stress on resistive switching mechanisms in hafnia-based ferroelectric tunnel junction

APPLIED PHYSICS LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwak, Been; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2023

Low-frequency noise characteristics of indium-gallium-zinc oxide ferroelectric thin-film transistors with metal-ferroelectric-metal-insulator-semiconductor structure

APPLIED PHYSICS LETTERS

Shin, Wonjun; Park, Eun Chan; Koo, Ryun-Han; Kwon, Dongseok; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 3

2023

Efficient Hybrid Training Method for Neuromorphic Hardware Using Analog Nonvolatile Memory

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Kwon, Dongseok; Woo, Sung Yun; Hwang, Joon; Kim, Hyeongsu; Bae, Jong-Ho; Shin, Wonjun; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 0

2023

Analog Synaptic Devices Based on IGZO Thin-Film Transistors with a Metal-Ferroelectric-Metal-Insulator-Semiconductor Structure for High-Performance Neuromorphic Systems

ADVANCED INTELLIGENT SYSTEMS

Kwon, Dongseok; Park, Eun Chan; Shin, Wonjun; Koo, Ryun-Han; Hwang, Joon; Bae, Jong-Ho; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2023

A novel pathway to construct gas concentration prediction model in real-world applications: Data augmentation; fast prediction; and interpolation and extrapolation

SENSORS AND ACTUATORS B-CHEMICAL

Kim, Jaehyeon; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Choi, Woo Young; Kim, Jae -Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 4

2023

1/f Noise in Synaptic Ferroelectric Tunnel Junction: Impact on Convolutional Neural Network

ADVANCED INTELLIGENT SYSTEMS

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 11

2023

Fast-response/recovery In₂O₃ thin-film transistor-type NO₂ gas sensor with floating-gate at low temperature

SENSORS AND ACTUATORS B-CHEMICAL

Yeom, Gyuho; Kwon, Dongseok; Shin, Wonjun; Park, Min-Kyu; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 10

2023

Low-frequency noise in gas sensors: A review

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong -Ho

被引用次数: 16

2023

Innovative Gas Sensing Method Using Transient Behavior of FET-type Sensors with Gate Pulse Input

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Jung, Gyuweon; Kim, Jaehyeon; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Lee, Chayoung; Choi, Woo Young; Lee, Jong-Ho

被引用次数: 0

2022

Effects of Electrode Structure on H₂S Sensing and Low-Frequency Noise Characteristics in In₂O₃-Based Resistor-Type Gas Sensors

IEEE SENSORS JOURNAL

Kim, Donghee; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Lee, Jong-Ho

被引用次数: 10

2022

Highly Efficient Self-Curing Method in MOSFET Using Parasitic Bipolar Junction Transistor

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Hong, Seongbin; Kwon, Dongseok; Hwang, Joon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 8

2022

Investigation of Low-Frequency Noise Characteristics of Ferroelectric Tunnel Junction: From Conduction Mechanism and Scaling Perspectives

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kwon, Dongseok; Koo, Ryun-Han; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 17

2022

Effects of Channel Length Scaling on the Signal-to-Noise Ratio in FET-Type Gas Sensor With Horizontal Floating-Gate

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 20

2022

Effects of High-Pressure Annealing on the Low-Frequency Noise Characteristics in Ferroelectric FET

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Sihyun; Lee, Kitae; Kwon, Dongseok; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 40

2022

Effects of Postdeposition Annealing Ambience on NO₂ Gas Sensing Performance in Si-Based FET-Type Gas Sensor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 10

2022

Variability analysis of ferroelectric FETs in program operation using low-frequency noise spectroscopy

APPLIED PHYSICS LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 12

2022

Synergistic improvement of sensing performance in ferroelectric transistor gas sensors using remnant polarization

MATERIALS HORIZONS

Shin, Wonjun; Yim, Jiyong; Bae, Jong-Ho; Lee, Jung-Kyu; Hong, Seongbin; Kim, Jaehyeon; Jeong, Yujeong; Kwon, Dongseok; Koo, Ryun-Han; Jung, Gyuweon; Han, Changhyeon; Kim, Jeonghan; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 19

2022

Highly Selective and Low-Power Carbon Monoxide Gas Sensor Based on the Chain Reaction of Oxygen and Carbon Monoxide to WO₃

ACS APPLIED MATERIALS & INTERFACES

Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Lee, Jong-Ho

被引用次数: 28

2022

Comprehensive and accurate analysis of the working principle in ferroelectric tunnel junctions using low-frequency noise spectroscopy

NANOSCALE

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Yim, Jiyong; Kwon, Dongseok; Kim, Yeonwoo; Yu, Junsu; Hwang, Joon; Park, Byung-Gook; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 29

2022

Fully integrated FET-type gas sensor with optimized signal-to-noise ratio for H₂S gas detection

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 17

2022

Optimization of channel structure and bias condition for signal-to-noise ratio improvement in Si-based FET-type gas sensor with horizontal floating-gate

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 17

2022

Efficient fusion of spiking neural networks and FET-type gas sensors for a fast and reliable artificial olfactory system

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Kim, Jaehyeon; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 26

2021

Gas sensing materials roadmap

JOURNAL OF PHYSICS-CONDENSED MATTER

Wang, Huaping; Ma, Jianmin; Zhang, Jun; Feng, Yuezhan; Vijjapu, Mani Teja; Yuvaraja, Saravanan; Surya, Sandeep G.; Salama, Khaled N.; Dong, Chengjun; Wang, Yude; Kuang, Qin; Tshabalala, Zamaswazi P.; Motaung, David E.; Liu, Xianghong; Yang, Junliang; Fu, Haitao; Yang, Xiaohong; An, Xizhong; Zhou, Shiqiang; Zi, Baoye; Liu, Qingju; Urso, Mario; Zhang, Bo; Akande, A. A.; Prasad, Arun K.; Hung, Chu Manh; Van Duy, Nguyen; Hoa, Nguyen Duc; Wu, Kaidi; Zhang, Chao; Kumar, Rahul; Kumar, Mahesh; Kim, Youngjun; Wu, Jin; Wu, Zixuan; Yang, Xing; Vanalakar, S. A.; Luo, Jingting; Kan, Hao; Li, Min; Jang, Ho Won;

Orlandi, Marcelo Ornaghi; Mirzaei, Ali; Kim, Hyoun Woo; Kim, Sang Sub; Uddin, A. S. M. Iftekhar; Wang, Jing; Xia, Yi; Wongchoosuk, Chatchawal; Nag, Anindya; Mukhopadhyay, Subhas; Saxena, Nupur; Kumar, Pragati; Do, Jing-Shan; Lee, Jong-Ho; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Bruzzi, Mara; Zhu, Chen; Gerald, Rex E., II; Huang, Jie

被引用次数: 56

2021

Effect of charge storage engineering on the NO₂ gas sensing properties of a WO₃ FET-type gas sensor with a horizontal floating-gate

NANOSCALE

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Lee, Chayoung; Park, Byung-Gook; Lee, Jong-Ho

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2021

Optimization of post-deposition annealing temperature for improved signal-to-noise ratio in In₂O₃ gas sensor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Park, Byung-Gook; Lee, Jong-Ho

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2021

Response Comparison of Resistor- and Si FET-Type Gas Sensors on the Same Substrate

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 4

2021

Effects of IGZO film thickness on H₂S gas sensing performance: Response, excessive recovery, low-frequency noise, and signal-to-noise ratio

SENSORS AND ACTUATORS B-CHEMICAL

Shin, Wonjun; Kwon, Daehee; Ryu, Minjeong; Kwon, Joowon; Hong, Seongbin; Jeong, Yujeong; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Lee, Jong-Ho

被引用次数: 24

2021

Highly stable Si MOSFET-type humidity sensor with ink-jet printed graphene quantum dots sensing layer

SENSORS AND ACTUATORS B-CHEMICAL

Jeong, Yujeong; Hong, Seongbin; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Choi, Yong Seok; Bae, Jong-Ho; Hong, Byung Hee; Lee, Jong-Ho

被引用次数: 15

2021

Low-power and reliable gas sensing system based on recurrent neural networks

SENSORS AND ACTUATORS B-CHEMICAL

Kwon, Dongseok; Jung, Gyuweon; Shin, Wonjun; Jeong, Yujeong; Hong, Seongbin; Oh, Seongbin; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 19

2021

Investigation of Low-Frequency Noise Characteristics in Gated Schottky Diodes

IEEE ELECTRON DEVICE LETTERS

Kwon, Dongseok; Shin, Wonjun; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho
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Impacts of Program/Erase Cycling on the Low-Frequency Noise Characteristics of Reconfigurable Gated Schottky Diodes

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Shin, Wonjun; Kwon, Dongseok; Bae, Jong-Ho; Lim, Suhwan; Park, Byung-Gook; Lee, Jong-Ho
被引用次数: 10 2021

Comparison of the characteristics of semiconductor gas sensors with different transducers fabricated on the same substrate

SENSORS AND ACTUATORS B-CHEMICAL

Jung, Gyuweon; Shin, Wonjun; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho
被引用次数: 35 2021

FET-type gas sensors: A review

SENSORS AND ACTUATORS B-CHEMICAL

Hong, Seongbin; Wu, Meile; Hong, Yoonki; Jeong, Yujeong; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Jang, Dongkyu; Lee, Jong-Ho
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A low-power embedded poly-Si micro-heater for gas sensor platform based on a FET transducer and its application for NO₂ sensing

SENSORS AND ACTUATORS B-CHEMICAL

Jung, Gyuweon; Hong, Yoonki; Hong, Seongbin; Jang, Dongkyu; Jeong, Yujeong; Shin, Wonjun; Park, Jinwoo; Kim, Donghee; Jeong, Chan Bae; Kim, Dong Uk; Chang, Ki Soo; Lee, Jong-Ho
被引用次数: 45 2021

Improved signal-to-noise-ratio of FET-type gas sensors using body bias control and embedded micro-heater

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Shin, Wonjun; Hong, Seongbin; Jung, Gyuweon; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Jang, Dongkyu; Park, Byung-Gook; Lee, Jong-Ho
被引用次数: 22 2021

Proposition of deposition and bias conditions for optimal signal-to-noise-ratio in resistor- and FET-type gas sensors

NANOSCALE

Shin, Wonjun; Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Kim, Donghee; Jang, Dongkyu; Kwon, Dongseok; Bae, Jong-Ho; Park, Byung-Gook; Lee, Jong-Ho
被引用次数: 34 2020

SO₂ gas sensing characteristics of FET- and resistor-type gas sensors having WO₃ as sensing material

SOLID-STATE ELECTRONICS

Jung, Gyuweon; Jeong, Yujeong; Hong, Yoonki; Wu, Meile; Hong, Seongbin; Shin, Wonjun; Park, Jinwoo; Jang, Dongkyu; Lee, Jong-Ho

被引用次数: 62

2020

Low frequency noise characteristics of resistor- and Si MOSFET-type gas sensors fabricated on the same Si wafer with In2O3 sensing layer

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Shin, Wonjun; Jung, Gyuweon; Hong, Seongbin; Jeong, Yujeong; Park, Jinwoo; Jang, Dongkyu; Park, Byung-Gook; Lee, Jong-Ho

被引用次数: 43

2020

Improved CO gas detection of Si MOSFET gas sensor with catalytic Pt decoration and pre-bias effect

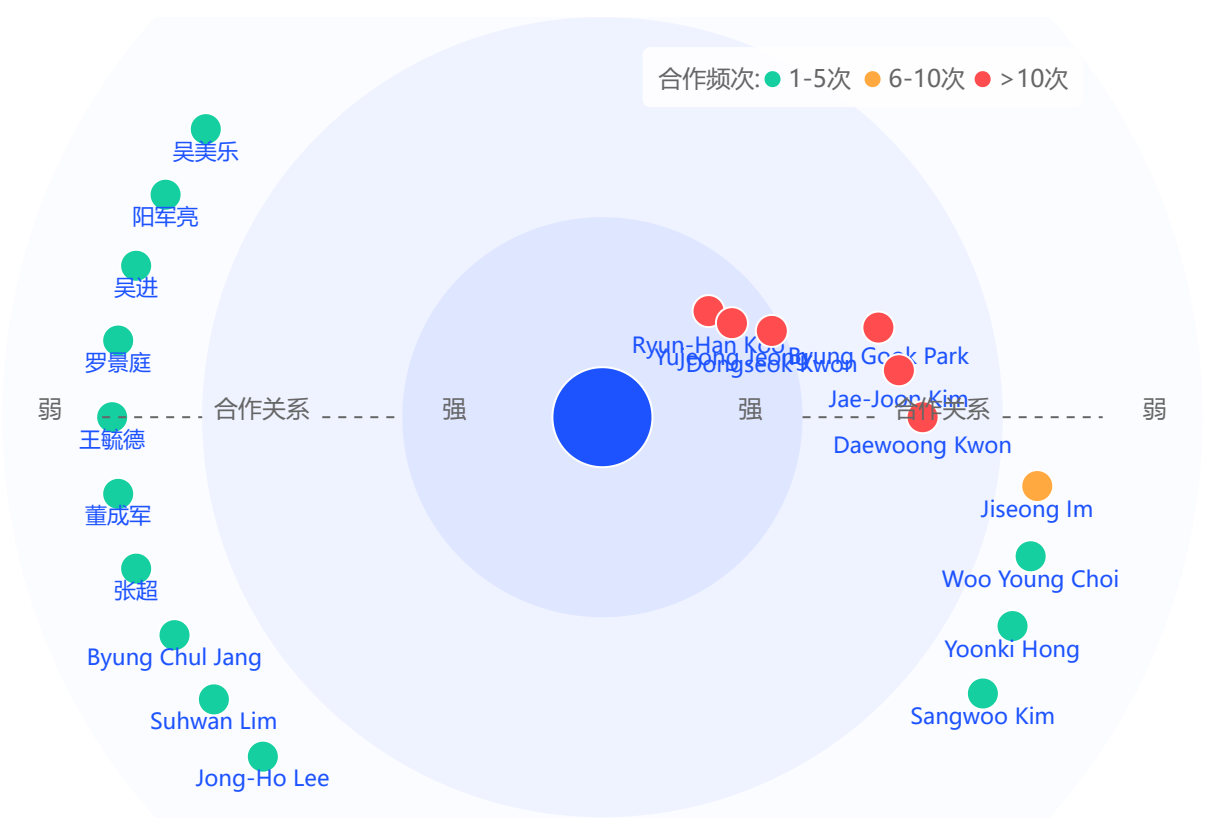
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Hong, Seongbin; Hong, Yoonki; Jeong, Yujeong; Jung, Gyuweon; Shin, Wonjun; Park, Jinwoo; Lee, Jung-Kyu; Jang, Dongkyu; Bae, Jong-Ho; Le, Jong-Ho

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